PROBLEM UNDERSTANDING

1. Introduction

Milk collection in rural areas, especially in Gujarat, is largely managed through manual processes. Farmers deliver milk to collection centres where quantity and quality are recorded in notebooks. This traditional approach often leads to inefficiencies, errors, and lack of transparency. A digital system is needed to streamline operations and improve reliability.

2. Stakeholders

- **Farmers:** Supply milk and receive payments
- **Collection Centre Staff:** Record milk data and manage daily operations
- **Cooperative Management:** Monitor performance and generate reports
- **System Administrators:** Maintain and manage the system

3. Problems Identified

- Manual data entry leads to errors
- Records can be lost or damaged
- Lack of transparency in payment calculation
- No real-time updates for farmers
- Difficulty in tracking historical data
- Inefficient reporting across multiple centres

4. Existing System & Its Limitations

The current system is paper-based and relies on manual recording. Quality testing is done using basic tools, and results are written down. Payment calculations are also done manually.

Limitations:

- Error-prone data recording

- No centralized data storage
- Time-consuming processes
- Lack of scalability
- Limited accessibility for farmers

5. Constraints & Assumptions

Constraints:

- Limited development time
- No paid tools or infrastructure
- Dependency on internet availability

Assumptions:

- Staff have basic digital literacy
- Collection centres have minimal device access
- Farmers rely on staff for data entry